許樂瑤 112700026

10. Marriage Rates Half of adults today are married compared with the all-time high of 72% in 1960.[9] Do marriage rates vary by educational level? A sample of 300 adults in each of three educational categories provided the following information.

DATA SET DS1406

	Married		
Educational Level	**Yes**	**No**	
Bachelors or higher	187	113	300
Some college	162	138	300
High school or less	149	151	300
	498	402	900

Test to determine whether marriage rates differ significantly among adults in these educational levels using $\alpha = .01$. How would you summarize your results?

H_0: Marital status and educational level are independent

H_a: Marital status and educational level are not independent

Education / Married	O_{ij}	$\hat{E}_{ij}$	$O_{ij}-\hat{E}_{ij}$	$\frac{(O_{ij}-\hat{E}_{ij})^2}{\hat{E}_{ij}}$
Bachelors or higher, Yes	187	166	21	2.6566
Some college, Yes	162	166	-4	0.0964
High School or less, Yes	149	166	-17	1.7410
Bachelors or higher, No	113	134	-21	3.2910
Some college, No	138	134	4	0.1194
High School or less, No	151	134	17	2.1567
				$\chi^2 = 10.0611$

(Yes) $\hat{E}_{ij} = \frac{(498)(300)}{900} = 166$

(No) $\hat{E}_{ij} = \frac{(402)(300)}{900} = 134$

$df = (r-1)(c-1) = (3-1)(2-1) = 2$

$\alpha = 0.01,\ df = 2$

$\chi^2_{2,0.01} = 9.2103$

$\because \chi^2 = 10.06 > 9.2103$

$\therefore$ Reject H_0

12. How Big Is the Household? A local chamber of commerce surveyed 120 households in their city—40 in each of three types of residence (apartment, duplex, or single residence)—and recorded the number of family members in each of the households. The data are shown in the table.

DATA SET DS1407

	Type of Residence			
Family Members	Apartment	Duplex	Single Residence	
1	8	20	1	29
2	16	8	9	33
3	10	10	14	34
4 or More	6	2	16	24
	40	40	40	120

Is there a significant difference in the family size distributions for the three types of residence? Test using $\alpha = .01$. If there are significant differences, describe the nature of these differences.

H_0: Family size and type of residence are independent

H_a: Family size and type of residence are not independent

FM / Type	O_{ij}	$\hat{E}_{ij}$	$O_{ij} - \hat{E}_{ij}$	$\frac{(O_{ij}-\hat{E}_{ij})^2}{\hat{E}_{ij}}$
1, A	8	$\frac{29(40)}{120} = 9.6667$	-1.6667	0.2874
2, A	16	$\frac{33(40)}{120} = 11$	5	2.2727
3, A	10	$\frac{34(40)}{120} = 11.3333$	-1.3333	0.1569
4 or more, A	6	$\frac{24(40)}{120} = 8$	-2	0.5
1, D	20	$\frac{29(40)}{120} = 9.6667$	10.3333	11.0459
2, D	8	$\frac{33(40)}{120} = 11$	-3	0.8182
3, D	10	$\frac{34(40)}{120} = 11.3333$	-1.3333	0.1569
4 or more, D	2	$\frac{24(40)}{120} = 8$	-6	4.5
1, SR	1	$\frac{29(40)}{120} = 9.6667$	-8.6667	7.7701
2, SR	9	$\frac{33(40)}{120} = 11$	-2	0.3636
3, SR	14	$\frac{34(40)}{120} = 11.3333$	2.6667	0.6275
4 or more, SR	16	$\frac{24(40)}{120} = 8$	8	8
				$\chi^2 = 36.4992$

$df = (r-1)(c-1)$

$= (4-1)(3-1) = 6$

$\alpha = 0.01,\ df = 6$

$\chi^2_{6,0.01} = 16.8119$

$\because \chi^2 = 36.4992 > 16.8119$

$\therefore$ Reject H_0

12. Dissolved Oxygen Content Dissolved oxygen content is a measure of the ability of a river, lake, or stream to support aquatic life, with high levels being better. A pollution-control inspector who suspected that a river community was releasing semitreated sewage into a river, randomly selected five specimens of river water at a location above the town and another five below. These are the dissolved oxygen readings (in parts per million):

Above Town	4.8	5.2	5.0	4.9	5.1
Below Town	5.0	4.7	4.9	4.8	4.9

a. Use a one-tailed Wilcoxon rank sum test with $\alpha = .05$ to confirm or refute the theory.

b. Use a Student's t-test (with $\alpha = .05$) to analyze the data. Compare the conclusion reached in part a.

(a) H_0 : Above and Below have the same distribution

H_a : Above Town has higher dissolved oxygen than Below Town

Value	4.7	4.8	4.8	4.9	4.9	4.9	5.0	5.0	5.1	5.2
Group	B	A	B	A	B	B	A	B	A	A
Rank	1	2.5	2.5	5	5	5	7.5	7.5	9	10

$T_A = 2.5 + 5 + 7.5 + 9 + 10 = 34$

$T_A^* = n_1(n_1+n_2+1) - T_A = 5(5+5+1) - 34 = 55 - 34 = 21$

$T = \min(T_A, T_A^*) = \min(34, 21) = 21$

$T_c = 19$

$\because T = 21 > T_c = 19$

$\therefore$ Non-reject H_0

(b) $\bar{x}_A = \dfrac{4.8+5.2+5.0+4.9+5.1}{5} = 5$

$\bar{x}_B = \dfrac{5.0+4.7+4.9+4.8+4.9}{5} = 4.86$

$s_1^2 = \dfrac{0.10}{5-1} = 0.025$

$s_2^2 = \dfrac{0.052}{5-1} = 0.013$

$\bar{x}_A - \bar{x}_B = 0.14$

$$s_p^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}$$

$$= \frac{4(0.025)+4(0.013)}{8}$$

$$= 0.019$$

$t_{8,0.05} = 1.86$

$\because t = 1.61 < t_{8,0.05} = 1.86$

$$t = \frac{\bar{x}_A - \bar{x}_B}{\sqrt{s_p^2(\frac{1}{n_A}+\frac{1}{n_B})}} = \frac{0.14}{\sqrt{0.0076}} = \frac{0.14}{0.08718} = 1.61$$

$\therefore$ Non-reject H_0

DATA SET DS1502

13. Eye Movement In an investigation of the visual scanning behavior of deaf children, measurements of eye movement were taken on nine deaf and nine hearing children. The table gives the eye-movement rates and their ranks (in parentheses). Does it appear that the distributions of eye-movement rates for deaf children and hearing children differ?

	Deaf Children	Hearing Children
	2.75 (15)	.89 (1)
	2.14 (11)	1.43 (7)
	3.23 (18)	1.06 (4)
	2.07 (10)	1.01 (3)
	2.49 (14)	.94 (2)
	2.18 (12)	1.79 (8)
	3.16 (17)	1.12 (5.5)
	2.93 (16)	2.01 (9)
	2.20 (13)	1.12 (5.5)
Rank Sum	126	45

H_0: The two distributions are the same

H_a: The two distributions are different

$n_1 = n_2 = 9$

$T = \min(T, T^*) = \min(45, 126) = 45$

$T_c = 63$

$\because T = 45 < T_c = 63$

$\therefore$ Reject H_0

Two Simple Examples *Use the sign test to compare two populations for significant differences for the paired data in Exercises 4–5. State the null and alternative hypotheses to be tested. Determine an appropriate rejection region with $\alpha \le .10$. Calculate the observed value of the test statistic and present your conclusion.*

4.

	Pair						
Population	1	2	3	4	5	6	7
1	8.9	8.1	9.3	7.7	10.4	8.3	7.4
2	8.8	7.4	9.0	7.8	9.9	8.1	6.9
	+	+	+	−	+	+	+

$n=7$, $x=6$

$H_0: p=0.5$
$H_a: p \neq 0.5$

$x \sim Bin(7, 0.5)$
$\because \alpha \le 0.10$, two-tailed

$P(x \le 1) = 0.062$ $\qquad$ $P(x \le 0) = 0.008$

p-value $= 2(0.062)$ $\qquad$ p-value $= 2(0.008)$
$= 0.124$ $\qquad$ $= 0.016$

$\therefore$ rejection region is $x=0$ or $x=7$

$\because x=6$, $RR = \{0 \le x \le 7\}$
$\therefore$ Non-reject H_0

11. Gourmet Cooking Two chefs, A and B, rated 22 meals on a scale of 1 – 10. The data are shown in the table. Do the data provide sufficient evidence to indicate that one of the chefs tends to give higher ratings than the other? Test by using the sign test with a value of α near .05.

DATA SET DS1507

Meal	A	B		Meal	A	B	
1	6	8	−	12	8	5	+
2	4	5	−	13	4	2	+
3	7	4	+	14	3	3	0
4	8	7	+	15	6	8	−
5	2	3	−	16	9	10	−
6	7	4	+	17	9	8	+
7	9	9	0	18	4	6	−
8	7	8	−	19	4	3	+
9	2	5	−	20	5	4	+
10	4	3	+	21	3	2	+
11	6	9	−	22	5	3	+

(−) = 9
(+) = 11

n = 20

x = 11

a. Use the binomial tables in Appendix I to find the exact rejection region for the test.

b. Use the large-sample z statistic. (NOTE: Although the large-sample approximation is suggested for $n \ge 25$, it works fairly well for values of n as small as 15.)

c. Compare the results of parts a and b.

$H_0: p = 0.5$, $H_a: p \neq 0.5$

(a) $X \sim Bin(20, 0.5)$

$P(X \le 5) = 0.0207$

$\because 20 - 5 = 15$

$\therefore X \ge 15$

$p\text{-value} = 2P(X \le 5)$

$= 2(0.0207)$

$= 0.0414$

$\because x = 11$, $RR = \{5 \le x \le 15\}$

$\therefore$ Non-reject H_0

(b) $z = \dfrac{x - \frac{n}{2}}{\sqrt{\frac{n}{4}}}$

$= \dfrac{11 - \frac{20}{2}}{\sqrt{\frac{20}{4}}}$

$= \dfrac{1}{\sqrt{5}}$

≈ 0.45

$z_{\frac{\alpha}{2}} = z_{0.025} = 1.96$

$\because z = 0.45 < z_{0.025} = 1.96$

$\therefore$ Non-reject H_0

(c) The results are the same

DATA SET DS1508 **13. Recovery Rates** Clinical data concerning the effectiveness of two drugs in treating a particular condition (as measured by recovery in 7 days or less) were collected from 10 hospitals. You want to know whether the data present sufficient evidence to indicate a higher recovery rate for one of the two drugs.

a. Test using the sign test. Choose your rejection region so that α is near .05.

b. Why might it be inappropriate to use the Student's t-test in analyzing the data?

	Drug A		
Hospital	Number in Group	Number Recovered (7 days or less)	Percentage Recovered
1	84	63	75.0
2	63	44	69.8
3	56	48	85.7
4	77	57	74.0
5	29	20	69.0
6	48	40	83.3
7	61	42	68.9
8	45	35	77.8
9	79	57	72.2
10	62	48	77.4

	Drug B			
Hospital	Number in Group	Number Recovered (7 days or less)	Percentage Recovered	
1	96	82	85.4	−
2	83	69	83.1	−
3	91	73	80.2	+
4	47	35	74.5	−
5	60	42	70.0	−
6	27	22	81.5	+
7	69	52	75.4	−
8	72	57	79.2	−
9	89	76	85.4	−
10	46	37	80.4	−

$(+) = 2$

$(-) = 8$

$n = 10$

(a) $H_0: p = 0.5$

$H_a: p \neq 0.5$

$X \sim Bin(10, 0.5)$

$n = 10, x = 2$

$P(X \leq 1) = 0.0107$

$\because 10 - 1$

$p\text{-value} = 2P(X \leq 1)$

$\therefore P(X \geq 9)$

$= 2(0.0107)$

$= 0.0214$

$\because x = 2, RR = \{1 \leq x \leq 9\}$

$\therefore$ Non-reject H_0

(b) The data are percentages and may not be normally distributed, so the sign test is more appropriate.